

DBMS: ASSIGNMENT 5

Cursor Program

Name: Ritobrata Baksi
PRN:124B1F152
Batch:A4

Case Study: Staff Bonus Processing & Tax Audit Suite

```
mysql> USE StaffBonusSystem;  
Database changed
```

```
mysql> CREATE TABLE Staff (  
-> Id INT PRIMARY KEY,  
-> Name VARCHAR(50),  
-> Salary INT,  
-> Score INT  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE bonus_log (  
-> log_id INT AUTO_INCREMENT PRIMARY KEY,  
-> Id INT,  
-> Action VARCHAR(30),  
-> old_salary INT,  
-> new_salary INT,  
-> log_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Staff VALUES  
-> (1,'Adi Maitre',10000,5),  
-> (2,'Kritika Chaudhari',15000,4),  
-> (3,'Ritobrata Baksi',20000,3),  
-> (4,'Sarang Kolekar',18000,2);  
Query OK, 4 rows affected (0.02 sec)
```

```
mysql> DELIMITER $$  
mysql> CREATE PROCEDURE ProcessBonusWithCursor()  
-> BEGIN  
-> DECLARE done INT DEFAULT 0;  
-> DECLARE v_id INT;  
-> DECLARE v_salary INT;
```

```

-> DECLARE v_score INT;
-> DECLARE v_action VARCHAR(30);

-> DECLARE staff_cursor CURSOR FOR
-> SELECT Id, Salary, Score FROM Staff;

-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

-> OPEN staff_cursor;

-> read_loop: LOOP
-> FETCH staff_cursor INTO v_id, v_salary, v_score;

-> IF done = 1 THEN
-> LEAVE read_loop;
-> END IF;

-> IF v_score = 5 THEN
-> SET v_salary = v_salary + 10000;
-> SET v_action = 'Outstanding';
-> ELSEIF v_score = 4 THEN
-> SET v_salary = v_salary + 5000;
-> SET v_action = 'Very Good';
-> ELSEIF v_score = 3 THEN
-> SET v_salary = v_salary + 2000;
-> SET v_action = 'Good';
-> ELSE
-> SET v_action = 'Needs Training';
-> END IF;

-> UPDATE Staff
-> SET Salary = v_salary
-> WHERE Id = v_id;

-> INSERT INTO bonus_log(Id, Action, old_salary, new_salary)
-> VALUES(v_id, v_action, v_salary, v_salary);

-> END LOOP;

-> CLOSE staff_cursor;
-> END$$
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;

```

```
mysql> CALL ProcessBonusWithCursor();
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SELECT * FROM Staff;
```

Id	Name	Salary	Score
1	Adi Maitre	20000	5
2	Kritika Chaudhari	20000	4
3	Ritobrata Baksi	22000	3
4	Sarang Kolekar	18000	2

```
mysql> SELECT * FROM bonus_log;
```

log_id	Id	Action	old_salary	new_salary	log_date
1	1	Outstanding	20000	20000	2026-04-14 11:28:26
2	2	Very Good	20000	20000	2026-04-14 11:28:26
3	3	Good	22000	22000	2026-04-14 11:28:26
4	4	Needs Training	18000	18000	2026-04-14 11:28:26

Project: Smart Tax Audit & Filing Suite

```
mysql> USE SmartTaxReturnSuite;
```

Database changed

```
mysql> INSERT INTO user_login (first_name,last_name,username,password_hash,role)
VALUES
```

```
-> ('Adi','Maitre','adi123','pass1','taxpayer'),
-> ('Kritika','Chaudhari','kritika123','pass2','taxpayer'),
-> ('Ritobrata','Baksi','rito123','pass3','taxpayer'),
-> ('Sarang','Kolekar','sarang123','pass4','taxpayer');
```

Query OK, 4 rows affected (0.01 sec)

```
mysql> INSERT INTO taxpayer_profile (user_id,pan_number,city,state) VALUES
```

```
-> (1,'PAN001','Pune','MH'),
-> (2,'PAN002','Mumbai','MH'),
-> (3,'PAN003','Delhi','DL'),
-> (4,'PAN004','Nashik','MH');
```

Query OK, 4 rows affected (0.01 sec)

```
mysql> INSERT INTO income_sources (taxpayer_id,source_type,amount,financial_year)
VALUES
```

```
-> (1,'Salary',600000,'2024-25'),  
-> (2,'Business',300000,'2024-25'),  
-> (3,'Salary',800000,'2024-25'),  
-> (4,'Freelance',200000,'2024-25');  
Query OK, 4 rows affected (0.01 sec)
```

```
mysql> INSERT INTO deductions (taxpayer_id,deduction_type,amount,financial_year) VALUES  
-> (1,'80C',50000,'2024-25'),  
-> (2,'80D',20000,'2024-25'),  
-> (3,'80C',60000,'2024-25'),  
-> (4,'80G',10000,'2024-25');  
Query OK, 4 rows affected (0.01 sec)
```

```
mysql> DELIMITER $$  
mysql> CREATE PROCEDURE TaxAuditCursor()  
-> BEGIN  
-> DECLARE done INT DEFAULT 0;  
-> DECLARE v_taxpayer BIGINT;  
-> DECLARE v_income DECIMAL(15,2);  
-> DECLARE v_tax DECIMAL(15,2);  
-> DECLARE total_tax DECIMAL(15,2) DEFAULT 0;  
  
-> DECLARE tax_cursor CURSOR FOR  
-> SELECT taxpayer_id, SUM(amount)  
-> FROM income_sources  
-> GROUP BY taxpayer_id;  
  
-> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;  
  
-> OPEN tax_cursor;  
  
-> read_loop: LOOP  
-> FETCH tax_cursor INTO v_taxpayer, v_income;  
  
-> IF done = 1 THEN  
-> LEAVE read_loop;  
-> END IF;  
  
-> IF v_income > 500000 THEN  
-> SET v_tax = v_income * 0.10;  
-> SET total_tax = total_tax + v_tax;  
  
-> INSERT INTO audit_logs(user_id, action, status, details)  
-> VALUES(v_taxpayer,'TAX_AUDIT','SUCCESS',
```

```
-> CONCAT('Tax: ', v_tax, ', Total: ', total_tax));  
-> END IF;
```

```
-> END LOOP;
```

```
-> CLOSE tax_cursor;  
-> END$$
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
```

```
mysql> CALL TaxAuditCursor();
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM audit_logs;
```

log_id	user_id	action	status	details
1	1	TAX_AUDIT	SUCCESS	Tax: 60000.00, Total: 60000
2	3	TAX_AUDIT	SUCCESS	Tax: 80000.00, Total: 140000